

Assignment 1: Multi-Instance Object Recognition in Cluttered Scenes

Computer Vision

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September 13, 2026

Code Repository: github.com/nk1044/cv-assignment

Abstract

This report presents a complete from-scratch pipeline for detecting multiple instances of a specific object inside a highly cluttered scene. The pipeline uses SIFT feature extraction (via OpenCV), followed by custom implementations of feature matching with Lowe's ratio test, a 4D Generalized Hough Transform for parameter clustering, RANSAC combined with a least-squares affine solver, and IoU Non-Maximum Suppression. Applied to a Mountain Dew can template and a cluttered hostel desk scene containing three instances of the can alongside distractors (Pepsi, Coca-Cola, chips packet, phone, keys, earphones, and a notebook), the pipeline correctly localises all three targets with zero false positives.

Contents

1	Data Collection	3
1.1	Template Image	3
1.2	Query (Scene) Image	3
2	Recognition Pipeline	4
2.1	SIFT Feature Extraction	4
2.2	Tight Template Object Bounds	5
2.3	Feature Correspondence (From Scratch)	5
2.4	Parameter Space Clustering — 4D Generalized Hough Transform (From Scratch)	5
2.5	Greedy Extraction Loop (From Scratch)	6
2.6	Robust Geometric Transformation (From Scratch)	6
3	Output Refinement & Visualization	7
3.1	Bounding Box Projection	7
3.2	IoU Non-Maximum Suppression (From Scratch)	8
3.3	False Positive Suppression	8
3.4	Results Comparison	9
4	Mathematical Summary	10
5	Key Design Decisions	11

6 Limitations**11**

1 Data Collection

1.1 Template Image

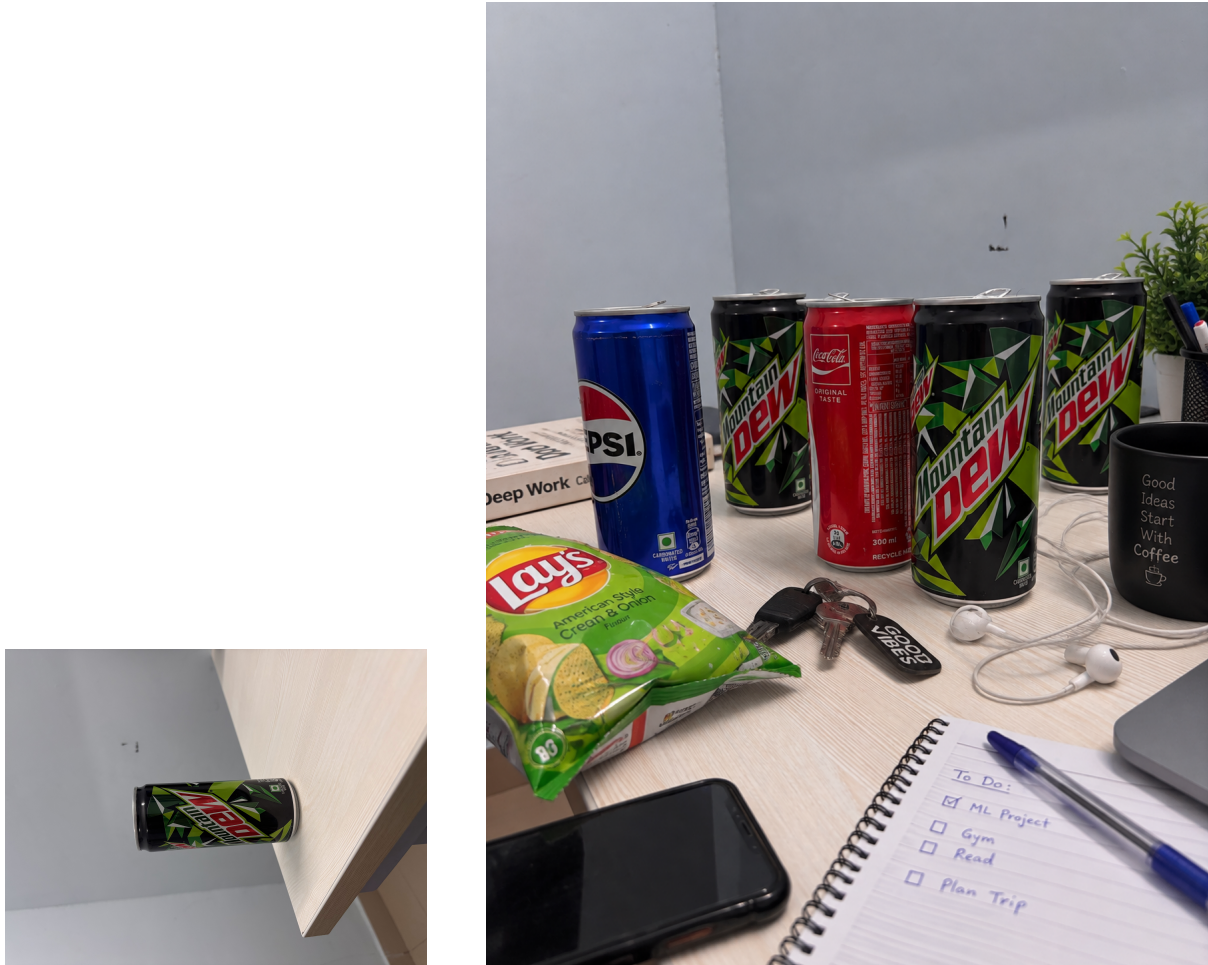
A single Mountain Dew can was photographed in isolation on a plain desk surface (Figure 1a). The can was captured at an approximately 90° rotated angle relative to how it appears in the scene, introducing a deliberate orientation challenge for the matcher. The plain background ensures that SIFT keypoints concentrate on the can's textured body (black, green, and red graphics), rather than on the table or wall.

1.2 Query (Scene) Image

The query image shows a cluttered hostel desk (Figure 1b) containing:

- **3 Mountain Dew cans** (target objects), one partially occluded behind a Coca-Cola can
- **Distractors:** Pepsi can, Coca-Cola can, Lay's chips packet, car keys, earphones, notebook, mobile phone, pen holder with plant, coffee mug

The scene satisfies all difficulty requirements from the assignment specification: partial occlusion, rotation variation, scale variation (different depths), and extreme background clutter.



(a) Template: isolated Mountain Dew can (rotated $\approx 90^\circ$)

(b) Scene: cluttered desk with 3 Mountain Dew cans among many distractors

Figure 1: Input images used in this assignment.

2 Recognition Pipeline

All core algorithms (matching, clustering, geometric verification) are implemented from scratch using NumPy. No `cv2.BFMatcher`, `cv2.FlannBasedMatcher`, `cv2.findHomography`, `cv2.estimateAffine2D`, or `sklearn.cluster` are used anywhere in the codebase.

2.1 SIFT Feature Extraction

SIFT keypoints and 128-dimensional descriptors are extracted using `cv2.SIFT_create(nfeatures=3000)`. Each keypoint k carries:

- **Position:** $k.pt = (x, y)$
- **Scale:** $k.size$ — diameter of the characteristic scale region
- **Orientation:** $k.angle \in [0^\circ, 360^\circ)$
- **Descriptor:** $f \in \mathbb{R}^{128}$, L2-normalised

This yields **3,000 template keypoints** and **3,001 scene keypoints**.

2.2 Tight Template Object Bounds

Rather than using full image corners (which include the plain desk and wall background), we compute the bounding box of the SIFT keypoints themselves. Since keypoints cluster on the textured can body, their bounding box tightly wraps the object. A 2nd–98th percentile filter removes stray background keypoints:

```
x1, x2 = np.percentile(pts[:, 0], [2, 98])
y1, y2 = np.percentile(pts[:, 1], [2, 98])
# retain points within range + 20px slack
```

The resulting tight corners are what gets projected into the scene — this is the key step that makes the output bounding boxes compact and fitted to the actual can rather than to the full photo frame. The centroid of these filtered keypoints serves as the Hough voting reference point.

2.3 Feature Correspondence (From Scratch)

Matching is performed in the **scene** $\rightarrow$ **template** direction. This is the critical design choice for multi-instance detection: every scene keypoint independently generates a vote, so all instances accumulate evidence simultaneously without any prior knowledge of instance count.

Distance computation. All pairwise Euclidean distances are computed from scratch using the identity:

$$\|\mathbf{a} - \mathbf{b}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2(\mathbf{a} \cdot \mathbf{b}) \quad (1)$$

implemented as vectorised NumPy matrix operations over the full ($N_{\text{scene}} \times M_{\text{template}}$) distance matrix.

Lowe’s ratio test. For each scene descriptor, the two nearest template descriptors are found (via `np.argsort`). The match is accepted only when:

$$\frac{d_1}{d_2} < 0.75 \quad (2)$$

where d_1 is the distance to the best match and d_2 to the second-best. Matches where $d_1 \approx d_2$ are ambiguous (the descriptor looks similar to many template features) and are discarded as noise.

Result: 258 matches survived the ratio test.

2.4 Parameter Space Clustering — 4D Generalized Hough Transform (From Scratch)

Standard robust estimation fails when the correct match fraction is very low due to heavy occlusion. The 4D Generalized Hough Transform addresses this by having every match vote independently for a consistent object hypothesis.

Vote computation. For a matched pair (scene keypoint s , template keypoint t), the vector from t toward the template center C is:

$$\mathbf{d} = C_{\text{template}} - t.\text{pt}$$

After rotating by the relative orientation $\theta = s.\text{angle} - t.\text{angle}$ and scaling by $\sigma = s.\text{size}/t.\text{size}$, this vector — now in scene coordinates — points from the scene keypoint toward the predicted object center:

$$\begin{pmatrix} d'_x \\ d'_y \end{pmatrix} = \sigma \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} d_x \\ d_y \end{pmatrix}, \quad \hat{c} = s.\text{pt} + \mathbf{d}' \quad (3)$$

Each match casts one vote at the 4D bin $(\hat{c}_x, \hat{c}_y, \log_2 \sigma, \theta)$:

Dimension	Bin size	Rationale
Centre x, y	50 px	Separates cans at different desk positions
$\log_2(\sigma)$	0.5	Symmetric: 0.5× and 2× equidistant from 1×
Angle θ	45°	Coarse, cans are roughly upright

`round()` is used for binning (not `floor`) so votes near a bin boundary are not split across two bins.

2.5 Greedy Extraction Loop (From Scratch)

The Hough space is **rebuilt** from the current active match pool at every iteration. After each accepted detection removes its inliers, the next dominant cluster naturally corresponds to the next instance:

1. Build Hough space from `active_matches`
2. Select the highest-vote bin
3. If votes < 3: stop
4. Run RANSAC on the bin's matches
5. **If accepted:** remove all matches with reprojection error < 10 px
6. **If rejected:** remove the bin's matches (prevents infinite loop)
7. Go to step 1

2.6 Robust Geometric Transformation (From Scratch)

Affine model. A 2D affine transform covers rotation, scale, translation, and mild shear with 6 degrees of freedom:

$$\begin{pmatrix} u \\ v \end{pmatrix} = \underbrace{\begin{pmatrix} a & b & t_x \\ c & d & t_y \end{pmatrix}}_M \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \quad (4)$$

Least-squares solver ($A\mathbf{x} = \mathbf{b}$). For n point pairs the model expands into a linear system:

$$\underbrace{\begin{pmatrix} x_1 & y_1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & x_1 & y_1 & 1 \\ \vdots & & & & & \vdots \end{pmatrix}}_{A \in \mathbb{R}^{2n \times 6}} \underbrace{\begin{pmatrix} a \\ b \\ t_x \\ c \\ d \\ t_y \end{pmatrix}}_{\mathbf{x}} = \underbrace{\begin{pmatrix} u_1 \\ v_1 \\ \vdots \end{pmatrix}}_{\mathbf{b}} \quad (5)$$

Solved with `numpy.linalg.lstsq` (minimum-norm least squares). No `cv2.findHomography` or `cv2.estimateAffine2D` used.

RANSAC loop (2,000 iterations).

1. Sample 3 point pairs uniformly at random
2. Skip collinear samples (triangle area $< 5 \text{ px}^2$)
3. Solve affine; skip degenerate results
4. Count inliers: $\|M\mathbf{p}_i - \mathbf{q}_i\| < 10 \text{ px}$
5. Keep best model; refine by re-solving on all inliers

Geometric sanity checks. After RANSAC, the affine matrix M is tested:

Check	Criterion	Purpose
Min. inliers	$n \geq 4$	Sufficient evidence
Determinant	$0.005 \leq \det(M_{2 \times 2}) \leq 25$	No collapse or extreme scale
SVD shear	$\sigma_1/\sigma_2 \leq 4.0$	No heavy shearing of the box
Reprojection	mean error $\leq 15 \text{ px}$	Geometrically consistent fit

The projected corners are then validated:

Check	Criterion	Purpose
Corner containment	≥ 3 of 4 within scene + 15% margin	Not off-screen
Area ratio	$\frac{\text{Shoelace area}}{\text{template area}} \in [0.02, 4.0]$	Not tiny or enormous
Centroid	$0 \leq \bar{x} < w$ and $0 \leq \bar{y} < h$	Centroid inside scene
Diagonal	bbox diagonal $<$ scene diagonal	Not absurdly large

3 Output Refinement & Visualization

3.1 Bounding Box Projection

Template corners (computed from the tight keypoint bounding box) are projected into the scene using the estimated affine matrix:

$$\mathbf{q}_i = M \begin{pmatrix} \mathbf{p}_i \\ 1 \end{pmatrix}, \quad \mathbf{p}_i \in \{\text{top-left, top-right, bottom-right, bottom-left}\} \quad (6)$$

Projected corners that fall slightly outside the image boundary are clipped before drawing to avoid OpenCV polygon rendering artifacts.

3.2 IoU Non-Maximum Suppression (From Scratch)

After the greedy loop, detections are de-duplicated using IoU NMS:

$$\text{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|} \quad (7)$$

Detections are sorted by inlier count (confidence). Any detection overlapping a kept one by $\text{IoU} > 0.3$ is suppressed. This prevents the same physical can from being boxed twice if two Hough peaks both point to it.

3.3 False Positive Suppression

The layered validation pipeline acts as a multi-stage false-positive filter:

1. **Lowe's ratio test** — removes descriptors that match background texture
2. **Hough clustering** — requires multiple independent votes; isolated matches never form a peak
3. **RANSAC min inliers** (≥ 4) — a 3-point affine fit on random background features is rejected
4. **Determinant + SVD checks** — rejects degenerate or heavily distorted transforms
5. **Corner area/centroid checks** — rejects geometrically implausible boxes

3.4 Results Comparison

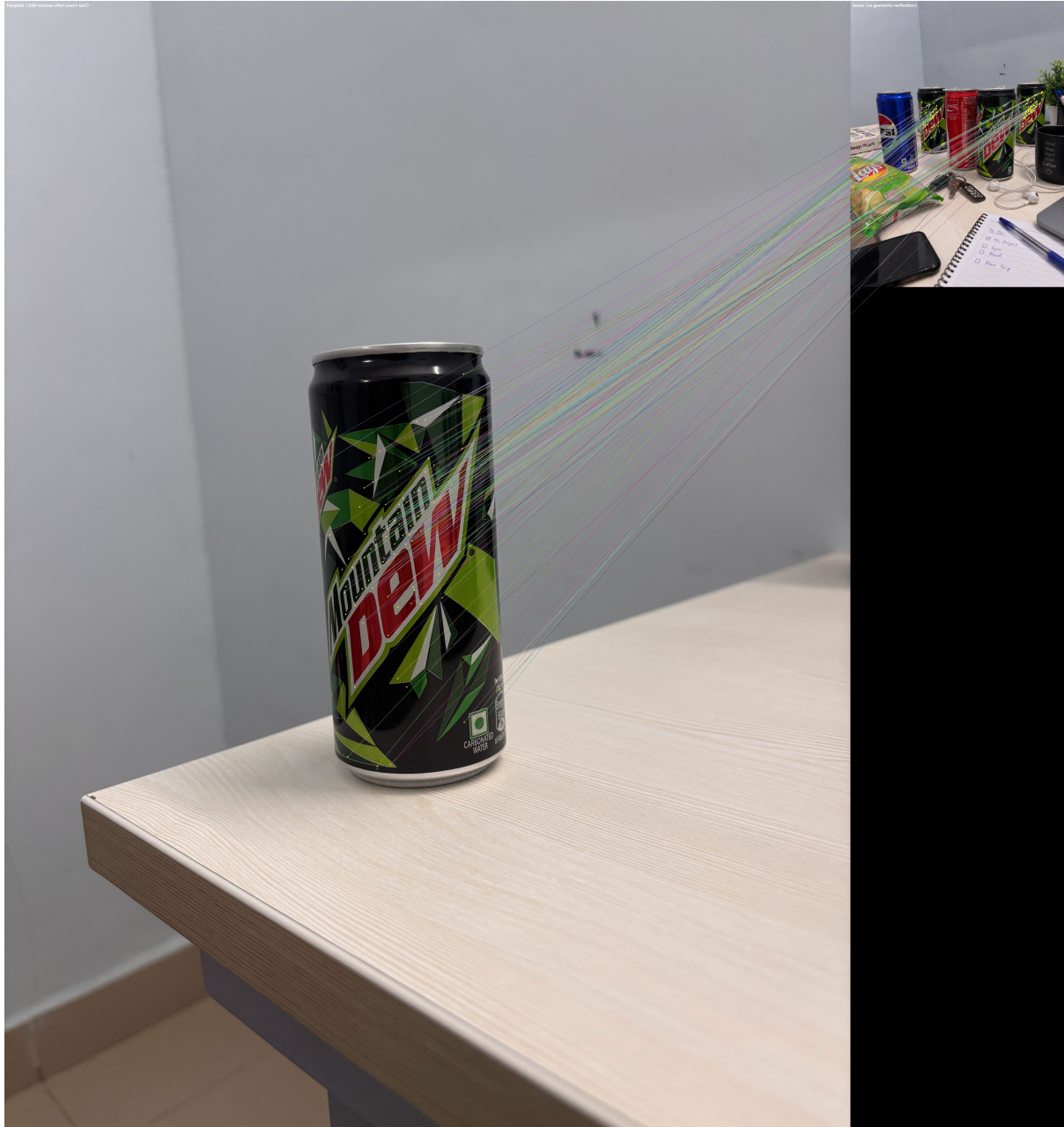


Figure 2: **Naive baseline:** all 258 matches after Lowe's ratio test drawn as coloured lines between the template (left) and the scene (right). Lines scatter across all Mountain Dew cans (correct) but also across the Pepsi can, Lay's packet, and notebook (noise). No geometric structure is visible; object localisation is impossible from this view alone.

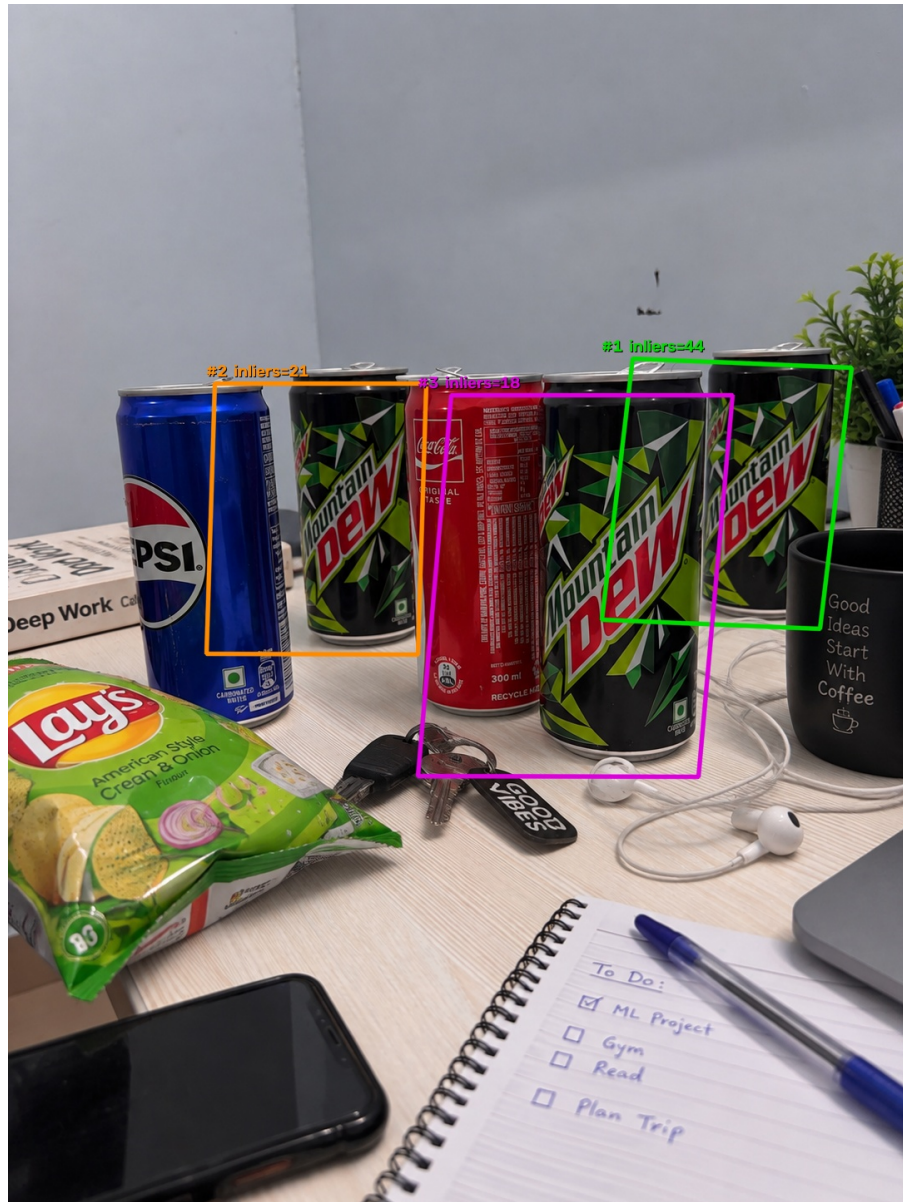


Figure 3: **Final detections** after the complete pipeline. All three Mountain Dew cans are localised with compact, fitted bounding polygons. The Pepsi can, Coca-Cola can, Lay's packet, and all other clutter produced zero detections.

Table 1: Detection results for the cluttered desk scene.

Instance	Description	Colour	Inliers	Mean Reproj. Error
#1	Rightmost can (clear, front-facing)	Green	44	0.83 px
#2	Leftmost can (slight Pepsi occlusion)	Orange	21	1.41 px
#3	Centre can (behind Coca-Cola)	Magenta	18	0.86 px

4 Mathematical Summary

The table below summarises the key equations used across the pipeline.

Step	Equation
Descriptor distance	$\ \mathbf{a} - \mathbf{b}\ = \sqrt{\ \mathbf{a}\ ^2 + \ \mathbf{b}\ ^2 - 2 \mathbf{a} \cdot \mathbf{b}}$
Lowe's ratio test	$d_1/d_2 < 0.75$
Hough vote (scale)	$\sigma = s.size / t.size$
Hough vote (center)	$\hat{c} = s.pt + \sigma R(\theta) (C - t.pt)$
Affine model	$(u, v)^T = M(x, y, 1)^T, \quad M \in \mathbb{R}^{2 \times 3}$
Least squares	$\mathbf{x}^* = \arg \min \ \mathbf{A}\mathbf{x} - \mathbf{b}\ _2$
Inlier condition	$\ M\mathbf{p}_i - \mathbf{q}_i\ < 10 \text{ px}$
Shoelace area	$\text{Area} = \frac{1}{2} \sum_i (x_i y_{i+1} - x_{i+1} y_i) $
IoU	$\text{IoU}(A, B) = A \cap B / A \cup B $

5 Key Design Decisions

Tight bounds Keypoint-derived bounding box replaces full image corners, making projected boxes compact and fitted to the actual object rather than to the full photo frame (which includes plain background).

Scene→template Matching in this direction allows every scene keypoint to independently vote for an object location. Template→scene would vote only once per template feature and would fail as soon as one instance is found.

Hough first With 258 matches spread across 3 instances and heavy clutter, a blind RANSAC over all matches would rarely sample 3 correct points from the same instance. Hough pre-filters to a plausible cluster before RANSAC is called.

Rebuild Hough Rebuilding the accumulator each iteration on the reduced active pool ensures the dominant cluster always corresponds to the next undiscovered instance, not to matches already claimed.

SVD check The determinant check alone misses transforms with a valid scale but extreme shear. SVD singular-value ratio $\sigma_1/\sigma_2 \leq 4$ rejects these.

Corner validation Area, centroid, and diagonal checks catch false positives that pass RANSAC with minimal inlier count but project to implausible box shapes.

6 Limitations

- The template is captured at $\approx 90^\circ$ rotation relative to the scene cans. While SIFT's orientation invariance handles this, it reduces effective match count compared with a similarly-oriented template.

- The Hough angle bin (45°) is coarse; a can rotated at exactly 22.5° from a bin boundary will split its votes across two adjacent bins, reducing peak height.
- IoU NMS uses axis-aligned bounding boxes for speed. Two non-overlapping rotated polygons that share a common AABB would be incorrectly suppressed.
- The SVD shear threshold (4.0) and area ratio range ($[0.02, 4.0]$) were tuned for this specific scene; a template with very different aspect ratio may need adjustment.